After Quiz 4, we speedily went through linear models. Let's recap quickly.

1 Introduction: predicting income levels

Linear models provide a mathematical relationship connecting an outcome, or response, to a bunch of predictors. There are many possible ways to relate a bunch of predictors to an outcome; the linear model provides one such way. (Other approaches are discussed in Section 3.)

Suppose you collect data on individual income levels (y). You believe that many factors influence how much an individual earns, for instance, job sector (academia, manufacturing, etc.) (x_1), education level (x_2), years of employment (x_3). You collect a bunch of such data — say p factors in total — for each individual whose income level you collected earlier. Your goal is to find a relationship connecting income levels to all the factors. There are many possible approaches, but suppose you believe that income levels should be *roughly* a linear combination of the other p factors,

$$y \approx x_1\beta_1 + \ldots + x_p\beta_p,$$

where the set of coefficients $\{\beta_i\}_{i=1}^p$ quantifies this linear combination and is *fixed no matter which individual you look at.*

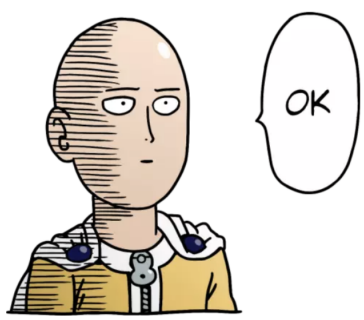

Figure 1: Your response to "$\{\beta_i\}_{i=1}^p$ is fixed no matter which individual you look at." Source: `https://shop.spreadshirt.com/specialtee-shop/`

Let's say you collected n samples. For each sample i, let y_i be the income level and $\{x_{i1}, \ldots, x_{ip}\}$ be the values measured for the p factors. Since you have n samples, you can summarise your data into a vector of outcomes (aka **response**) $\mathbf{y}$ and a matrix of factors (aka **predictors** or **covariates**) $\mathbf{X}$.

$$\mathbf{y} = \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix}, \qquad \mathbf{X} = \begin{pmatrix} x_{11} & x_{12} & \cdots & x_{1p} \\ x_{21} & x_{22} & \cdots & x_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \cdots & x_{np} \end{pmatrix}$$

Then your belief about the linear relationship is equivalent to you believing that you can find $\{\beta_i\}_{i=1}^p$ such that

$$\begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix} \approx \begin{pmatrix} x_{11} & x_{12} & \cdots & x_{1p} \\ x_{21} & x_{22} & \cdots & x_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \cdots & x_{np} \end{pmatrix} \begin{pmatrix} \beta_1 \\ \vdots \\ \beta_p \end{pmatrix}. \tag{1}$$

Letting $\boldsymbol{\beta} = (\beta_1, \ldots, \beta_p)^T \in \mathbb{R}^{p\times 1}$ (the transpose of a row vector is a column vector), eq. (1) is equivalent to $\mathbf{y} \approx \mathbf{X}\boldsymbol{\beta}$.[1]

One immediate issue is that you probably won't find $\{\beta_i\}_{i=1}^p$ that achieves an *exact fit*, i.e., makes $\mathbf{y} = \mathbf{X}\boldsymbol{\beta}$ (note the equality!) owing to inexactness of the model. To resolve the issue while staying true to your earlier belief, you decide that randomness can account for the lack of exact fit or the roughness. There are many kinds of random effects possible, but if you believe that randomness distorts the linear relationship equally positively and negatively on average (i.e., **unbiased**), has same variability from individual to individual (i.e., **same variance**), and is normally distributed, then your model looks like

$$y_i = x_{i1}\beta_1 + \ldots + x_{ip}\beta_p + \varepsilon_i, \qquad \varepsilon_i \overset{\text{i.i.d.}}{\sim} N(0, \sigma^2)$$

for each $i \in \{1, \ldots, n\}$. If you summarise the n equations, you'd obtain

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}, \qquad \boldsymbol{\varepsilon} \sim \text{MVN}(\mathbf{0}, \sigma^2 I_n), \tag{2}$$

where MVN denotes a multivariate normal distribution which you will not be tested on but is nevertheless useful knowledge. (Note that $\boldsymbol{\varepsilon} = (\varepsilon_1, \ldots, \varepsilon_n)^T$ and I_n is the $n \times n$ identity matrix.)

If you followed through each sentence and line of reasoning in the preceding paragraphs, then congratulations, you've developed a linear model for your data!

> In general, if you collect a size n sample and each observation consists of an outcome y_i and a bunch of p predictors $\{x_{i1}, \ldots, x_{ip}\}$, then eq. (2) is a linear model for your data.

1.1 More on assumptions

As mentioned in today's lab, the assumption of mean zero equal variance normality of noise (ε_i) has many implications. We mention a few of them here.

1. Distribution of $\mathbf{y}$

   From eq. (2), it immediately follows that

   $$\mathbf{y} \sim \text{MVN}\left(\mathbf{X}\boldsymbol{\beta}, \sigma^2 I_n\right). \tag{3}$$

   In particular, as a **random vector**, $\mathbf{y}$ has expectation $\mathbb{E}[\mathbf{y}] = \mathbf{X}\boldsymbol{\beta}$ and covariance $\text{Cov}(\mathbf{y}) = \sigma^2 I_n$. The **covariance of a random vector** is a generalisation of the variance of a random variable: if $\mathbf{Z} = (Z_1, \ldots, Z_n)^T$ is a n-dimensional random vector, then $\text{Cov}(\mathbf{Z})$ is a $n\times n$ matrix, with (i, j)th entry equal to the covariance of random variables Z_i, Z_j, i.e., $\text{Cov}(Z_i, Z_j)$. This is just a more abstract way of saying that the y's are independently normally distributed: $y_i \overset{\text{i.i.d.}}{\sim} N(x_{i1}\beta_1 + \ldots + x_{ip}\beta_p, \sigma^2)$. Random vectors and covariance matrices are discussed in pp. 567-570 of Rice.

2. Homoscedasticity

   By the previous point, we observe that the linear model's assumption implies the variance of each response y_i is the same ($= \sigma^2$). This means that, if we fit the regression line $\mathbf{y} = \mathbf{X}\hat{\boldsymbol{\beta}}$

[1] If you feel shaky about this matrix equation, read about matrix multiplication on Wikipedia, or come to office hours.

to the sample, we should expect the **residuals** e_i, i.e., the vertical distance between the observed and the predicted values of y, to be equally distributed about zero and have similar magnitudes. Thus, to check whether the equal variance assumption of the noise terms ε_i is satisfied, oftentimes a scatterplot of y against x is generated and inspected (assuming there is only one predictor). The idea is that, if the equal variance assumption holds, then you can fit a straight line through the cloud of points such that the residuals from the points to this line are more or less equally distributed above and below, and don't display any trend in magnitude. Looking at Figure 2, can you try to fit a straight line to the left and the right clouds of points such that the residuals are more or less equally distributed and don't display trends in magnitude?

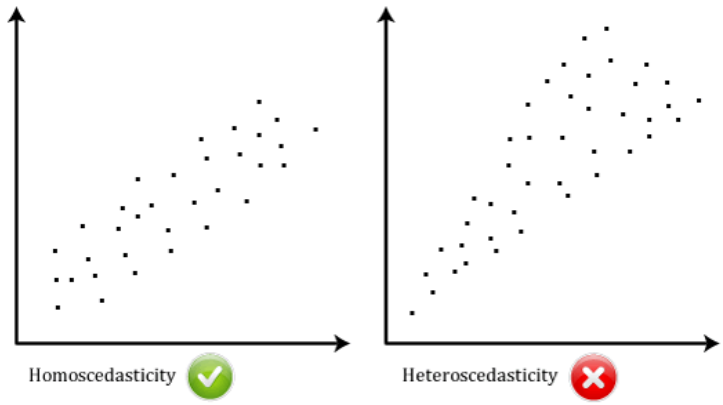

Figure 2: Homoscedasticity versus heteroscedasticity. Source: `https://stats.stackexchange.com/questions/76151/what-is-an-intuitive-explanation-of-why-we-want-homoskedasticity-in-a-regression/76181`

1.2 Why use linear models?

As mentioned in today's lab, linear models are convenient and mathematically wieldy, meaning that we know unbiased estimators for parameters of interest, and have concrete and exact ways of quantifying their uncertainty, including hypothesis tests and confidence interval construction. We will see these claims hold in Section 2.

Q1. Ok, so how will I use linear models outside this class?

Linear models (and more general models described in Section 3) are, collectively, *useful prediction machines*.[2] In the field of **statistical learning**, you wish to train a machine to look at a bunch of predictors and subsequently churn out a prediction. In our earlier example, you might want to predict how much an individual's income level is, based on the predictors you collected earlier on. A more telling example is to predict risks of diseases using biological predictors (e.g., gene expression levels). Such a prediction relies on a *learned function* f that relates the predictors $x = (x_1, \ldots, x_p)^T$ to the response y: $y = f(x)$. The learned function f is itself determined by your choice of model (a neural network, a support vector machine, a linear model, etc.), and f is learned from training data, i.e., the parameters of the model are

[2] I borrowed this term from the lucidly written book by Ajay Agrawal, Avi Goldfarb, and Joshua Gans. I highly recommend you guys read it if you're interested in learning broadly how machine learning can be integrated into companies, organisations and society.

estimated from training data. After learning f, you receive new data (just the predictors x, no response y!) and use your learned f to predict the response y.

The linear model corresponds to the function $f(x) = x^T\hat{\beta}$, where $\hat{\beta}$ is estimated from training data. If the linear model actually is a good model for the data you're looking at, then you'd be able to profit from fitting the linear model onto your training data and using the model to make predictions on new data.

Q2. I feel lukewarm about statistical learning. Perhaps I'm more interested in applications to social sciences. Are linear models still relevant?

Yes they are. Many social science papers that don't do predictions, still rely on linear models because of their convenience and mathematical wieldiness. Quite often, social scientists apply linear models without appreciating the assumptions behind them. (There are also social scientists who are very meticulous about this; look up Kosuke Imai who wrote a great textbook on quantitative social science.) They might not be aware that by fitting a linear model to their data and performing statistical tests downstream, they are assuming that whatever outcome they are measuring roughly has a linear relationship with the predictors. If this assumption is not true in reality, then their confidence intervals, hypotheses rejections, p-values, etc., are all invalid!

2 Important Facts

(Most of what I am writing here can be found in Chapter 14 of Rice.)

After prescribing a linear model, we'd like to know how to *estimate* $\boldsymbol{\beta}$, which relates the covariates to the response. Recall that the entries of $\boldsymbol{\beta}$ provide information about the effect of each predictor on the response.

The estimate of $\boldsymbol{\beta}$ in the linear model is

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}.$$

This formula can be implemented in **R** via the line

```
beta_hat ← solve(t(X) %*% X) %*% t(X) %*% y.
```

More tips on matrix computation in **R** are available here.

We verify that $\hat{\boldsymbol{\beta}}$ is unbiased:

$$\mathbb{E}[\hat{\boldsymbol{\beta}}] = \mathbb{E}\left[(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}\right] = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbb{E}[\mathbf{y}] \overset{\text{by (3)}}{=} \cancel{(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{X}}\boldsymbol{\beta} = \boldsymbol{\beta}.$$

How did we get the formula for $\hat{\boldsymbol{\beta}}$? As Adam mentioned in Lecture 32, $\hat{\boldsymbol{\beta}}$ minimises the **residual sum of squares (RSS)**, which is defined by

$$\text{RSS} = ||\mathbf{y} - \hat{\mathbf{y}}||_2^2 = ||\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}||_2^2 = \sum_{i=1}^{n}(y_i - x_{i1}\hat{\beta}_1 - \ldots - x_{ip}\hat{\beta}_p)^2. \tag{4}$$

So $\hat{\boldsymbol{\beta}}$ is the solution to the problem

$$\min_{\mathbf{c}\in\mathbb{R}^{p\times 1}} ||\mathbf{y} - \mathbf{X}\mathbf{c}||_2^2.$$

There are other good properties about $\hat{\boldsymbol{\beta}}$ (e.g., **best linear unbiasedness**) that we won't discuss.

In the previous box we wrote the formula for $\hat{\boldsymbol{\beta}}$. There, we implicitly assumed that $\mathbf{X}^T\mathbf{X}$ is invertible. This is not true in general. To make this true, we enforce the assumptions

$$n > p \text{ and } \mathbf{X} \text{ is full rank, meaning that } \operatorname{rank}(\mathbf{X}) = p.$$

Rice, on p. 566, shows that the assumption above is equivalent to $\mathbf{X}^T\mathbf{X}$ being invertible.

Now that we know how to estimate $\hat{\boldsymbol{\beta}}$, we also want to be able to *quantify the uncertainty* of our estimate.

Under the linear model,

$$\operatorname{Cov}(\hat{\boldsymbol{\beta}}) = \sigma^2(\mathbf{X}^T\mathbf{X})^{-1}. \tag{5}$$

Moreover, because σ^2 is unknown, we use the plugin estimator

$$\hat{\sigma^2} = \frac{\text{RSS}}{n-p}, \tag{6}$$

where RSS was defined in eq. (4).

To establish (5), observe that since $\hat{\boldsymbol{\beta}} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}$ and $\mathbf{y} \sim \text{MVN}\left(\mathbf{X}\boldsymbol{\beta}, \sigma^2 I_n\right)$, a fact about affine transformations of multivariate normal distributions[3] guarantees that $\hat{\boldsymbol{\beta}}$ also follows a MVN, and the same fact also shows how to compute the covariance matrix of the MVN:

$$\begin{aligned}
\operatorname{Cov}(\hat{\boldsymbol{\beta}}) &= (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\operatorname{Cov}(\mathbf{y})((\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T)^T \\
&= (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\sigma^2 I_n\mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1} \\
&= \sigma^2\cancel{(\mathbf{X}^T\mathbf{X})^{-1}}\cancel{\mathbf{X}^T\mathbf{X}}(\mathbf{X}^T\mathbf{X})^{-1} \\
&= \sigma^2(\mathbf{X}^T\mathbf{X})^{-1}.
\end{aligned}$$

Thus,

$$\hat{\boldsymbol{\beta}} \sim \text{MVN}\left(\boldsymbol{\beta}, \sigma^2(\mathbf{X}^T\mathbf{X})^{-1}\right) \tag{7}$$

I leave you to review p. 576 (Theorem A) of Rice, which states that $\hat{\sigma^2}$ as defined in eq. (6) (they use the notation s^2) is unbiased.

Knowing that $\hat{\boldsymbol{\beta}}$ is MVN (see eq. (7)) is extremely helpful. Since $\operatorname{Cov}(\hat{\boldsymbol{\beta}})$ is just the matrix with (i,j)th entry equal to $\operatorname{Cov}(\hat{\beta}_i, \hat{\beta}_j)$ and (i,i)th diagonal entry equal to $\operatorname{Var}(\hat{\beta}_i) = s^2_{\hat{\beta}_i}$, the distribution of $\hat{\beta}_i$ is normal:

$$\hat{\beta}_i \sim N(\beta_i, [\sigma^2(\mathbf{X}^T\mathbf{X})^{-1}]_{ii}) = N(\beta_i, s^2_{\hat{\beta}_i}).$$

This shows that $(\hat{\beta}_i - \beta_i)/s_{\hat{\beta}_i}$ has the standard $N(0,1)$ distribution. BUT REMEMBER: because we don't know σ^2 and plug $\hat{\sigma^2}$ as its estimate when computing $\operatorname{Cov}(\hat{\boldsymbol{\beta}})$, we actually end up computing an *estimate* of $\operatorname{Var}(\hat{\beta}_i)$. Just like how we've seen that $(\overline{X} - \mu_{\overline{X}})/\sigma$, which is typically normally distributed, becomes t distributed as soon as we replace σ with an unbiased estimator $s_{\overline{X}}$, we end

[3] See the Section on "Affine Transformation" in this Wikipedia article about MVNs.

up with a t distribution instead of a normal distribution when we account for this additional plugin estimation step.

In the linear model, $(\hat{\beta}_i - \beta_i)/s_{\hat{\beta}_i}$, where $s_{\hat{\beta}_i}$ is computed from the estimated covariance matrix $\hat{\sigma^2}(\mathbf{X}^T\mathbf{X})^{-1}$, is t distributed. Specifically,

$$\frac{\hat{\beta}_i - \beta_i}{s_{\hat{\beta}_i}} \sim t_{n-p},$$

where n is the number of samples and p is the number of estimated parameters (i.e., the size of the set $\{\beta_i\}_{i=1}^p$).

This last fact about the distribution of the normalised $\hat{\beta}_i$ coefficients can be used to construct confidence intervals: a $100(1-\alpha)\%$ confidence interval for β_i is

$$\hat{\beta}_i \pm t_{n-p}(\alpha/2)s_{\hat{\beta}_i}.$$

More broadly, from eq. (7) you can also determine distributions of *linear combinations of* $\hat{\beta}_i$*'s*. For instance, to get the distribution of $\hat{\beta}_1 - \hat{\beta}_2 = (1, -1, 0, \ldots, 0) \cdot \hat{\boldsymbol{\beta}}$, a fact about linear transformations of MVNs shows that

$$\hat{\beta}_1 - \hat{\beta}_2 \sim N\left(\beta_1 - \beta_2, (1, -1, 0, \ldots, 0)\sigma^2(\mathbf{X}^T\mathbf{X})^{-1}\begin{pmatrix} 1 \\ -1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}\right) = N(\beta_1 - \beta_2, \mathrm{Var}(\hat{\beta}_1) + \mathrm{Var}(\hat{\beta}_2) - 2\mathrm{Cov}(\hat{\beta}_1, \hat{\beta}_2)).$$

As before, because we are estimating $\sigma^2(\mathbf{X}^T\mathbf{X})^{-1}$ by $\hat{\sigma^2}(\mathbf{X}^T\mathbf{X})^{-1}$, when you normalise $\hat{\beta}_1 - \hat{\beta}_2$ you will end up with a t distribution. The degrees of freedom is still $n - p$, owing to p parameters being estimated.

2.1 Implementation in R

This is extremely straightforward. Given your data `y` and `X`, combine them (using, say, `cbind()`) into a data frame `mydata`, and you can fit the linear model in one single line:

```
my.model ← lm(y ~ x1 + x2, data=mydata)
```

Nothing beats learning by practising and following examples; here are some good sources.

3 Extensions

We mentioned that linear models provide one way to relate a bunch of predictors to an outcome. A large community of statisticians, and neighbouring communities of computer scientists, computational biologists, and econometricians, amongst others, share a history of investigating statistical models of which the linear model is the iconic example. These investigations include (1) why would this particular statistical model be valid? What evidence from the field supports the relationship between outcome and predictors as described by the model? (2) not paying attention to whether

the statistical model makes epistemological sense, can the model at least deliver good predictions? How accurately can the model predict?

If you delve deeper into the field, you will find that lines are blurry, and occasionally you might find yourself developing models that address not one of the two questions exclusively but the twain.[4] Of course, the ideal scenario is that you find a model that accords well with the actual mechanism relating the predictors and the outcome, and predicts well additionally.[5] But such "success stories" are rare, and if anything, even some well-established scientific theories about mechanisms linking predictors to outcomes are being challenged by the rise of huge data sets together with our ability to analyse them.[6] This might explain why statisticians still have great job prospects in a wide variety of fields.

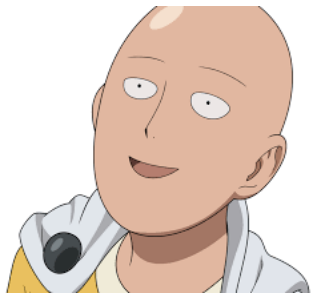

Figure 3: Your reaction to "why statisticians still have great job prospects in a wide variety of fields." Source: `https://steemit.com/anime/@samdaman/anime-to-watch-one-puch-man`

Below, I list some directions of pursuit that extend the theory of linear models covered in Stat 135.

1. How flexible is the linear model?

   Pretty flexible. You can actually transform your predictors and fit a linear model on the transformed predictors instead. For example, if you believe that the relationship between income level and years of employment is not linear but quadratic, then you could fit a linear model $y = x\beta_1 + x^2\beta_2$, where x is the number of years of employment. Note that even though this model is quadratic in x, it is still linear in the predictors x and x^2, and fitting a linear model to predictors x and x^2 will provide you the least squares estimates $\hat{\beta}_1$ and $\hat{\beta}_2$. Implementing this in **R** is straightforward, you can find an answer by Googling.

   You can also include **interaction terms** between predictors. For example, you believe that income level is not affected by a linear combination of education level x_1 and years of em-

[4] An example is Gaussian Process modeling for dynamic processes, which makes fairly minimal assumptions about the function relating predictors to outcomes.

[5] Such scenarios abound in physics. I am not an expert in physics, but you can look up statistical mechanics and see how researchers who study physically motivated models perform experiments that validate these models.

[6] For example, before epigenetics it was thought that physiological differences between humans could be entirely explained by differences in our genetic codes and mechanisms thereof. However, when scientists started looking at patterns of epigenetic variation (epigenetic processes are influenced by the environment and modify gene expression levels), they found new differences, revealing that it's not just genes but other biological processes owing to the environment that determine human physiological differences.

ployment x_2 but possibly a multiplicative combination of both[7]. Then you could fit a linear model with a term x_1x_2: $y = x_1\beta_1 + x_2\beta_2 + x_1x_2\beta_3$. This is straightforward to implement in **R**, Google will tell you how.

2. What happens when the linear model is not actually reasonable?

   You could try to develop your own model. A better approach is to consider already available models out there that account for non-linearity. A good book to look at is *Introduction to Statistical Learning* by Daniela Witten et al., which describes a library of models and methods tested on real data sets. The book has clear exposition and assumes few mathematical prerequisites. If you want to challenge yourself, there is a more theoretical version of this book, called *Elements of Statistical Learning*.

3. What happens when $n << p$, i.e., you have many more predictors than observations?

   Recall that $n > p$ is an important assumption for $\hat{\boldsymbol{\beta}}$ to be well-defined. So when this issue pops up and you still wish to fit a linear model on all the predictors, you should probably perform **variable selection**. Variable selection methods include the **LASSO** and the more general **elastic net**. Such data sets are very common in various fields.

[7] Such multiplicative effects of predictors actually exist in the biological context of epistasis, see this article to get some idea.